



Informed Consent Form

Development of AI-based System for Detection and Classification of Pronunciation Errors in the Holy Quran

Name of Principal Investigator: Faza Amal Sophian

Name of Organization: Department of Mechatronics Engineering International Islamic University Malaysia

This Informed Consent Form has two parts:

Part I: Information Sheet (to share information about the research study with you)

Part II: Certificate of Consent (for signatures if you agree to participate)

PART I: INFORMATION SHEET

Introduction

You are invited to participate in a research study conducted by Faza Amal Sophian from the Department of Mechatronics Engineering, International Islamic University Malaysia.

This study aims to collect audio recordings of Qur'ānic recitation to support the development of an AI system that can detect pronunciation errors, particularly in difficult Arabic letters.

Purpose of the Study

The purpose of this research is to develop a system capable of identifying mispronunciations in Qur'ānic recitation, focusing on letters such as 'Ayn (ع) and Dād (ض).

The study will contribute to improving digital tools for learning correct Qur'ānic pronunciation (Tajwīd), especially for non-Arab learners.

Type of Participation

Your participation will involve:

1. Reading and pronouncing specific Arabic letters shown on a screen
2. Recording your voice for different pronunciation variations

3. Reciting a short Qur'ānic passage
4. Uploading your recordings through an online form

The session will take approximately **10–15 minutes**.

Voluntary Participation

Your participation in this study is entirely voluntary. You may choose to withdraw at any time without any penalty or negative consequences.

Benefits

While there is no direct financial reward, your participation will:

- Contribute to research benefiting Qur'ānic education
- Help improve AI tools for Tajwīd learning

Risks and Discomforts

This study poses **minimal to no risk**. You may experience slight discomfort while recording your voice, but you may stop at any time if needed.

Confidentiality

All information collected will be kept strictly confidential:

- Your identity will **not be linked** to your recordings
- Data will be anonymized using participant IDs
- Recordings will only be used for research purposes

Data Usage

Your recordings will be used for:

- Training and evaluating machine learning models
- Academic research and publications (without identifying you)

Who to Contact

If you have any questions regarding this study, please contact:

Faza Amal Sophian
fazasophian123@gmail.com

PART II: CERTIFICATE OF CONSENT

I, _____, have read and understood the information provided above. I have had the opportunity to ask questions and all my enquiries have been answered to my satisfaction. I consent voluntarily to participate in this study.

Participant's Signature

Date:

(Day/Month/Year)